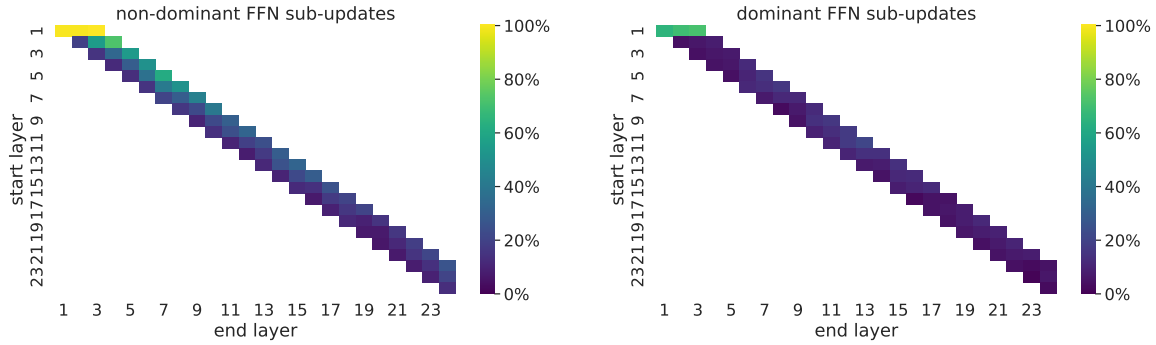


Figure 3: Intervention in non-dominant (left) and dominant (right) FFN sub-updates in $\text{GPT2}_{\text{M}}$. Each cell shows the percentage of memorized idioms for which the prediction was changed by zeroing-out the FFN sub-updates between the start and end layers.

ful for studying memory recall, and to test layers' roles in that process. Concretely, we zero out FFN sub-updates to the hidden representation (§4) and measure changes in memorized predictions.

**A Short Primer on Transformer FFN Sublayers.** FFN sublayers are the final computation in transformer layers, which output the hidden distribution at the center of our analysis. In general, they have a key role in capturing knowledge in transformer LMs (Dai et al., 2022; Meng et al., 2022). We follow Geva et al. (2022) and view the computation of each FFN sublayer as a weighted collection of $d_m$ sub-updates to the output distribution, each promoting a concept in the vocabulary space, e.g. "past-tense verbs" or "female athletes". To understand this, consider the computation of the FFN at layer $\ell$, given by $\text{FFN}^\ell(\mathbf{h}_i^\ell) = f(W_K^\ell \mathbf{h}_i^\ell) W_V^\ell$, where $W_K^\ell, W_V^\ell \in \mathbb{R}^{d_m \times d}$ are parameter matrices, $d_m$ is the intermediate hidden dimension, and $f$ is a point-wise non-linearity activation function. This computation can be decomposed as: $\text{FFN}^\ell(\mathbf{h}_i^\ell) = \sum_{j=1}^{d_m} f(\mathbf{h}_i^\ell \cdot \mathbf{k}_j^\ell)\mathbf{v}_j^\ell = \sum_{j=1}^{d_m} m_j^\ell \mathbf{v}_j^\ell$, where $\mathbf{k}_j^\ell$ and $\mathbf{v}_j^\ell$ are the $j$-th row in $W_K^\ell$ and the $j$-th column in $W_V^\ell$, respectively. Geva et al. argue that each weight $m_j^\ell$ is the score assigned by the model for some textual pattern, and each vector $\mathbf{v}_j$ promotes a concept that follows that pattern.

**Experiment** First, we sample 100 random instances from IDIOMEM that the model memorized. Then, for each range of up to 3 consecutive layers, we perform two complementary experiments, where we run $\text{GPT2}_{\text{M}}$'s inference on the 100 instances while intervening in the chosen layers to cancel the contribution of FFN sub-updates to the prediction. Specifically, for each layer $\ell$ in the layer range, we perform the following two interventions: first, we zero out (i.e. artificially set to 0 during inference) the weights of the 10 most dominant sub-updates, which are known to be particularly salient for predictions (Geva et al., 2022) (there are $d_m$ sub-updates per layer). Concretely, we sort the sub-updates by their contribution to the FFN output $|m_i^\ell| ||\mathbf{v}_i^\ell|| \; \forall i \in [1, ..., d_m]$, and set $m_i^\ell = 0$ for the 10 sub-updates with the highest contribution. Next, we zero out non-dominant sub-updates, i.e. all the sub-updates *except* for the 10 most dominant ones. For each intervention, we measure how often it changes the predicted token. Further measurements of the change in rank and probability of the target token are reported in §C.

**Results** Fig. 3 shows, for each layer range, the percentage of memorized idioms where the predicted token has changed. Focusing on zeroing out non-dominant FFN sub-updates (Fig. 3, left), we observe a *two-phase pattern of decreasing "layer importance" which corresponds to the two-phase pattern of decreasing rank and increasing probability during inference (§5.2)*: layers' effect on memory recall drops precipitously in the first 10 layers, corresponding to the candidate promotion phase. Then, from around layer 10 onwards, the drop in effect is much less steep, corresponding to the confidence-boosting phase. Intervention in upper layers rarely changes the predicted token, and its effect is limited to reducing the model's confidence (§C). We further visualize this two-phase behavior in Fig. 7 in §C.

These findings suggest that the candidate promotion phase, while having a smaller effect on the prediction's assigned probability compared to the later confidence-boosting stage, in fact, has a crucial role in memory recall.

We also observe that interventions in the first

layer are by far the most destructive, with 100% change in prediction for non-dominant updates (Fig. 3, left). Unlike for other layers, this is also observable when zeroing out the dominant sub-updates (Fig. 3, right), which constitute only 0.1% of layer sub-updates. This suggests the first layer is especially critical for memorized predictions.

## 6 Memorization of Factual Statements

We now examine if our findings generalize beyond idioms to other types of memory recall, focusing on the completion of statements expressing facts.

**Data** Datasets for evaluating memorization of factual knowledge typically contain simple queries such as *"The continent of Kuwait is"*, where predicting the next token requires knowledge of the triplet $\langle s, r, t \rangle$ where $s$ is a source entity (e.g., `Kuwait`), $t$ a target entity (e.g., `Asia`), and $r$ is the relation between them (e.g., `is the continent of`). However, unlike idioms, such queries are not suitable for probing memorization since they often fail to satisfy the criteria in §2. Concretely, queries often include "clues" that could make the prediction easy to guess and based on generalization (Poerner et al., 2020) (e.g. predicting a Spanish-speaking country for the query *"Federico López was born in"*), and the same fact can be expressed in multiple different ways (e.g. *"Kuwait is a country in Western Asia"* also encodes the above fact).

To test memorization of facts, we, therefore, collect factual statements where such cases are excluded. We use LAMA-UHN (Poerner et al., 2020), a subset of the LAMA dataset (Petroni et al., 2019), where "easy-to-guess" queries are filtered out. LAMA comprises of queries structured as "fill-in-the-blank" cloze statements (e.g. *"Gordon Scholes is a member of the ____political party."*). To accommodate autoregressive LMs, we consider only queries where the blank appears at the end. In addition, we keep only queries with a single correct completion (based on LAMA). This turns our definition of memorized and non-memorized sets (§4) equivalent to separating based on the evaluation metric of LAMA: an instance is considered memorized if the model predicts the single correct completion and non-memorized otherwise. Overall, the resulting collection consists of 17,855 factual statements with 22 unique relations.

**Memorized Facts Exhibit a Similar Prediction Process to Idioms** We repeat our analysis

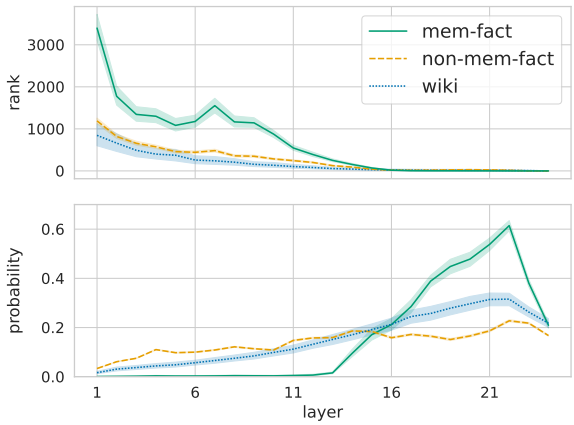


Figure 4: The predicted token's probability and rank across layers of $\text{GPT2}_\text{M}$, for memorized (`mem-fact`) and non-memorized (`non-mem-fact`) facts and short sequences from Wikipedia (`wiki`).

(§5.1), using the collected factual statements and $\text{GPT2}_\text{M}$. Splitting the statements to those memorized and non-memorized by the model results in 786 `mem-facts` vs. 17,069 `non-mem-facts` statements.

Fig. 4 shows the rank and probability of the predicted token across layers. Like idioms, memorized facts exhibit a clear two-phase prediction process, where the prediction probability rapidly increases once the candidate reaches a low rank (at layer 16). This is in contrast to non-memorized facts and short sequences from Wikipedia, where the rank (probability) gradually decreases (increases) across layers without a distinct two phases.

**Differences from Memorized Idioms Stem from Ill-Defined Targets** There is one major difference compared to memorized idioms (Fig. 4 vs. Fig. 2 upper left), which is a substantial drop in probability ($0.62 \rightarrow 0.21$) in the last two layers. We hypothesize that this is because the input query has multiple plausible completions that were not specified as "correct" targets in LAMA. We verify this by manually analyzing predictions, and find that for 82 out of 100 queries, there is more than one correct continuation in the top 5 predicted tokens. We posit that the above deficiency of LAMA is inherent because, in violation of our criterion (§2), factual statements can usually be expressed in many ways so their prompt has no single correct target.

## 7 Related Work

**Memorization as Training-Data Influence.** Memorization in LMs has attracted immense attention due to their rich sensitive training data (Carlini et al., 2019; Song and Shmatikov, 2019; Carlini et al., 2021; Zhang et al., 2021b; Carlini et al., 2022; Tirumala et al., 2022; Tänzer et al., 2022; Raunak and Menezes, 2022). Recent work suggests that memorization is *necessary* for performant ML due to the "long tail" of infrequently-observed patterns (Zhang et al., 2021a; Feldman, 2020; Brown et al., 2021; Raunak et al., 2021). This line of work has two key limitations: (a) only black-box behavior is measured rather than looking at models' internal prediction process, and (b) it detects memorized instances by measuring the effect of their inclusion in the training set on inference behavior. This results in a set of memorized examples that is specific to the model, training data, and even training pass, making it difficult to build on these results in future research.

**Transformers and Idioms.** Nedumpozhimana and Kelleher (2021) showed that idioms are identified using textual cues within the expression; Dankers et al. (2022) showed that idioms tend to be internally processed as single units of meaning. It has also been known (Fakharian and Cook, 2021; Salton et al., 2016) that LM contextual embeddings encode information about whether or not an expression is idiomatic (vs. literal). (Shwartz and Dagan, 2019; Chakrabarty et al., 2022) also studied representations and interpretation of non-compositional sequences, such as idioms. No prior work used idioms to probe LM memorization, which is one of the main contributions of this work. We release our dataset, IDIOMEM, to facilitate future research on memorization recall in LMs. Diagnostic datasets, such as IDIOMEM, have often proven useful in the past (Sugawara et al., 2022; Parrish et al., 2021).

**Memorization of Factual Knowledge.** An extensive line of work (Petroni et al., 2019; Jiang et al., 2020; Poerner et al., 2019; Lewis et al., 2020; Elazar et al., 2021) studied LMs' capacity to acquire relational knowledge during training. Some attention has also been given to understanding the inner workings of factual-memory recall: Wallat et al. (2020) showed that some facts are retrieved from the bottom and intermediate layers, and Meng et al. (2022) localized factual-knowledge recall within the feed-forward-component computation. Since factual statements do not fulfill our criteria, it is difficult to convincingly argue that correct predictions indicate memory recall, making it impossible to use them to isolate the effect of memorization.

## 8 Conclusion

We introduce a methodological framework for detecting and analyzing memorized predictions in transformer LMs. This includes a set of criteria on textual sequences for probing memorized predictions, the IDIOMEM dataset of idioms fulfilling these criteria, and an interpretation method of prediction internals. We characterize a behavioral profile that is unique to predictions involving memory recall and is observable across different LMs and data types. By providing these fundamental tools and initiating a thread of research on the phenomena we observe, we hope to empower future work toward demystifying transformer memorization.

## Limitations

Our criteria for detecting memorized instances are sufficient but not necessary, which raises the question of what other sequences that trigger memory recall satisfy them.

Additionally, while correct prediction for sequences that fulfill our criteria implies memory recall, incorrect prediction does not necessarily imply that no memory was used. This means that our set `non-mem-idiom` might include some portion of memorized sequences. This does not qualitatively affect our results as long as `mem-idiom` still contains *more* memorized predictions than `non-mem-idiom`, as is evidenced by the stark differences we observe between LMs' internal behaviors on these sets. Our work focuses on showing the utility of idioms for probing memorization, and opening up a new thread of research in this vein. We therefore leave further investigation of the above gaps for future work.

## Acknowledgements

We thank Gabriel Stanovsky for helpful feedback. This project has received funding and resources from CIFAR through the Canada CIFAR AI Chair program, the Province of Ontario, companies sponsoring the Vector Institute, as well as the Computer Science Scholarship granted by the Séphora Berrebi Foundation, and the European Research